Firas Naber

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Skills

Languages: Python, R, Java, SQL, C, C++
Frameworks & Libraries: Pandas, NumPy, scikit-learn, Matplotlib, Spring Boot, Hibernate
Tools & Systems: PySpark, GCP, Git, UNIX/Linux, gRPC

Experience

Graduate Research Assistant, University of Tulsa – Tulsa, OK — Jan 2025 – Dec 2025

- Processed and analyzed large-scale healthcare data from 278M individuals using SQL and Python (Pandas, NumPy)
- Automated the generation of 3K logistic regression models to predict disease risk, improving modeling pipeline efficiency
- Implemented 10K Cox proportional hazards models in R to perform survival analysis across multiple clinical datasets

Data Scientist, Laureate Institute for Brain Research – Tulsa, OK — Nov 2022 – Jan 2025

- Built and applied SEM, GLMs, and random forest models to analyze behavioral and clinical data from 200K subjects
- Automated preprocessing pipelines for large-scale genomic data using Python and SQL, reducing data cleaning time by 60%
- Designed and executed 9 longitudinal studies examining associations between brain structure and mental health outcomes
- Developed ETL pipelines using PySpark on GCP, processing and warehousing 138M records for downstream analysis

Software Engineer Intern, Eagle Creek – Vermillion, SD — June 2022 – Aug 2022

- Developed a recommendation system backend using Java, Spring Boot, and Hibernate with RESTful APIs serving 30K users
- Optimized database query performance in PostgreSQL using indexing strategies, reducing response times by 15%
- Wrote unit and integration tests using JUnit to validate API functionality and improve system reliability

Projects

gRPC-Based Distributed File System — Code Available on Request

- Engineered a distributed file system using gRPC streaming RPCs to support efficient, chunk-based file transfer across a client–server architecture
- Designed a multi-client synchronization protocol using CRC checksums and timestamp-based conflict resolution to ensure data consistency
- Implemented asynchronous callback handling with completion queues to coordinate client updates without blocking
- Built thread-safe concurrency control (mutexes, condition variables, server-side write locks) to handle simultaneous client operations

Multithreaded HTTP Proxy with Shared-Memory Cache — Code Available on Request

- Developed a high-performance multithreaded HTTP proxy using pthreads and libcurl to concurrently fetch and stream content to multiple clients
- Designed and implemented an IPC-based caching layer using POSIX shared memory, message queues, and semaphores for low-latency interprocess communication
- Engineered a chunked data transfer pipeline via mmap-backed shared memory to minimize redundant data copying between processes
- Built a synchronized shared-memory segment pool to safely manage concurrent requests and prevent race conditions

Education

Georgia Institute of Technology – MS in Computer Science — Expected: Aug 2028
University of Minnesota – BA in Computer Science and Statistics — May 2022